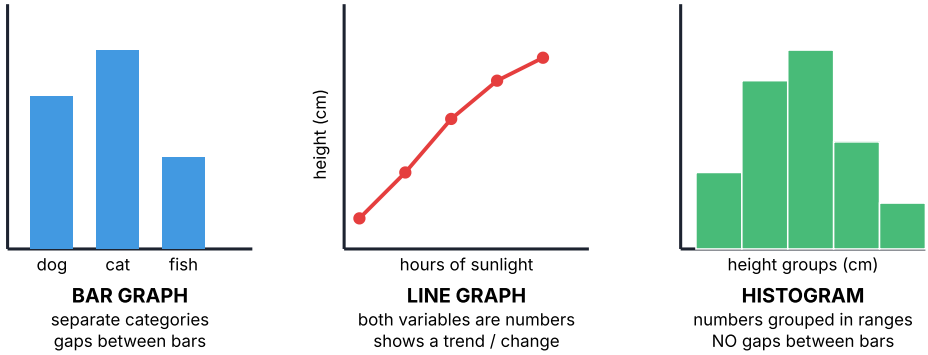
Grade 9 Natural Sciences (Biology paper) – Summary

The key facts and diagrams on a few pages. Use your full notes for the detailed explanations.

Scientific investigation

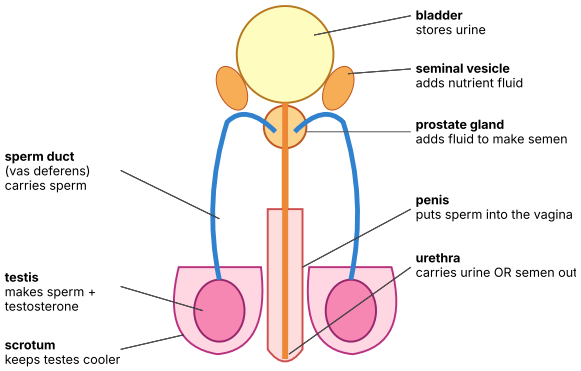
- **Hypothesis:** a testable prediction – "If [I change ...], then [... will happen]".
- **Independent variable:** what you change. **Dependent variable:** what you measure. **Controlled variables:** what you keep the same.
- **Reliable** = repeatable (repeat and average). **Valid** = a fair test (only one variable changed).
- **Bar graph:** categories, gaps. **Line graph:** both variables numbers, shows a trend. **Histogram:** grouped ranges, no gaps.
- **Graph rules:** heading; independent variable on the x-axis, dependent on the y-axis; label axes with units; even scale.



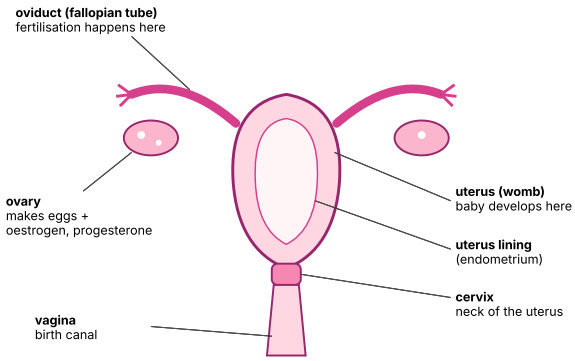
The three types of graphs

Human reproduction

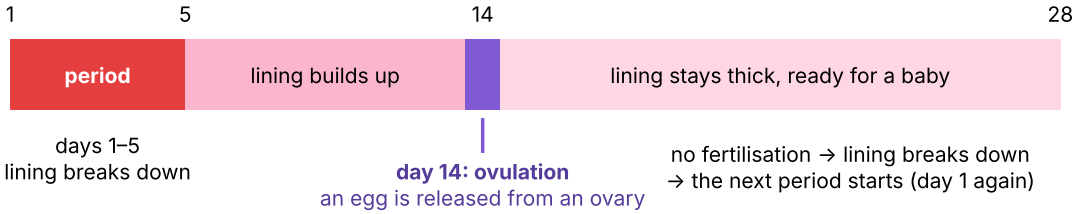
- **Puberty** (about 9–16 years): hormones testosterone (boys) and oestrogen/progesterone (girls). Growth spurt, body hair, voice deepens / breasts and hips develop, periods start.
- **Male:** testes make sperm and testosterone → sperm duct → seminal vesicle and prostate add fluid (semen) → urethra → penis. Scrotum keeps testes cool.
- **Female:** ovaries make eggs and hormones → oviduct (fertilisation happens here) → uterus (baby grows) → cervix → vagina.



The male reproductive system (front view, simplified)



The female reproductive system (front view, simplified)

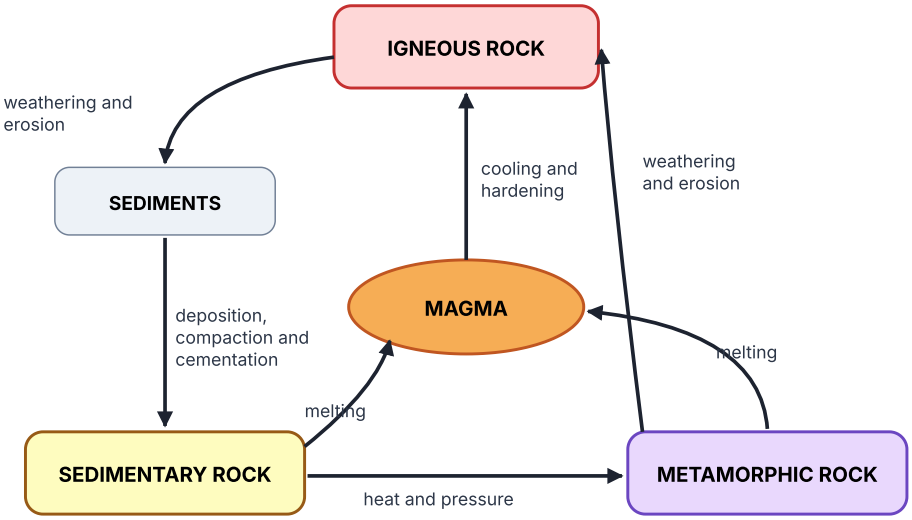


The 28-day menstrual cycle

- **Menstrual cycle** (28 days): days 1–5 period (lining breaks down) → lining thickens → **day 14 ovulation** (egg released) → lining stays thick → no fertilisation = next period.
- **Fertilisation:** sperm + egg in the oviduct → zygote → divides → **implants** in the uterus lining → embryo → foetus → birth after about 9 months.
- **Placenta:** food and oxygen pass from mother to baby, waste from baby to mother; joined by the umbilical cord. Alcohol, smoking and drugs can harm the baby.
- **Prevention:** abstinence (100%), condoms (also protect against STIs/HIV), the pill/injection (no STI protection).

The Earth as a system

- **Atmosphere** (air), **hydrosphere** (all water), **lithosphere** (crust + upper mantle), **biosphere** (all living things). They all interact – e.g. plants take CO_2 from the air (bio ↔ atmo); rain erodes rock (hydro → litho); volcanoes release gas (litho → atmo).
- **Earth's layers:** crust → mantle → liquid outer core → solid inner core (iron and nickel).



Rocks keep changing from one type to another over millions of years – there is no start or end.

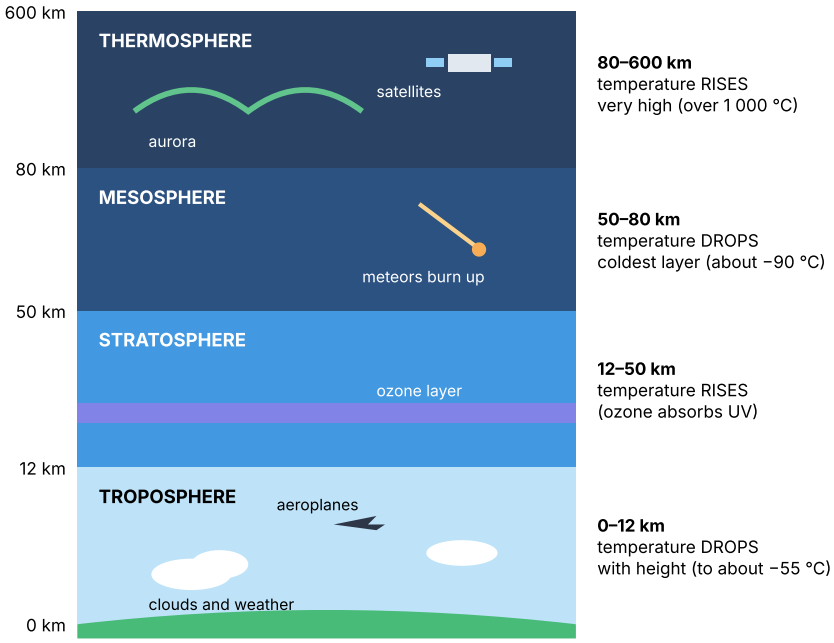
The rock cycle

Rock	Forms by	Examples
Igneous	Magma/lava cools and hardens	Granite, basalt
Sedimentary	Sediments deposited in layers, compacted and cemented	Sandstone, shale, coal
Metamorphic	Heat and pressure change existing rock	Marble, slate

Mining

- **Mineral:** natural, non-living solid with a definite composition. **Ore:** rock with enough mineral to mine profitably. Minerals are **non-renewable**.
- **Extraction:** open-cast (near the surface – coal, iron, diamonds); underground shaft (deep – gold, platinum); alluvial (rivers/sea – diamonds).
- **From ore to metal:** mine → crush → separate (concentrate) → **refine** (smelting, chemicals e.g. cyanide for gold, electrolysis) → pure metal.
- **SA:** gold (Witwatersrand), platinum (Bushveld), coal (Mpumalanga), diamonds (Kimberley), iron ore (Sishen).
- **Good:** jobs, exports, economy. **Bad:** land damage, acid mine drainage, dust/air pollution, dangerous work, minerals run out. Land must be **rehabilitated**.

The atmosphere

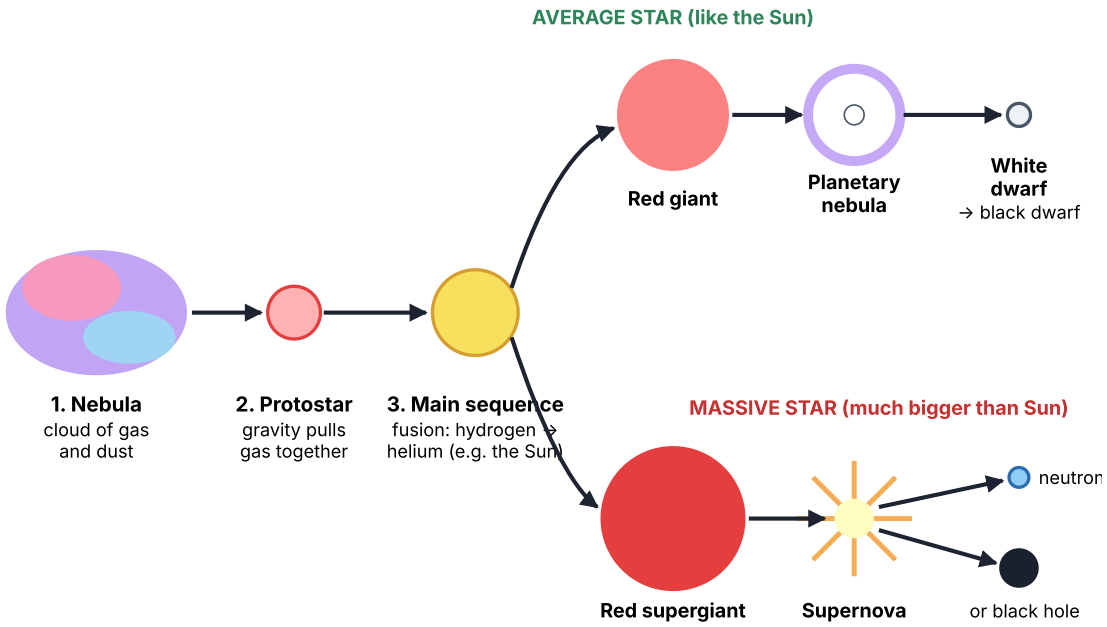


The four layers of the atmosphere (not to scale)

Layer	Height	Temperature going up	Key fact
Troposphere	0–12 km	Decreases	All weather; most of the air
Stratosphere	12–50 km	Increases	Ozone layer absorbs UV
Mesosphere	50–80 km	Decreases (coldest)	Meteors burn up
Thermosphere	80–600 km	Increases (very hot)	Aurora, satellites

- **Greenhouse effect:** sunlight warms the Earth → Earth gives off heat → greenhouse gases (CO_2, methane, water vapour) trap some heat. Naturally good – keeps Earth at about 15 °C.
- **Enhanced greenhouse effect → global warming:** burning fossil fuels and deforestation add extra CO_2 → rising sea levels, droughts, floods, extinctions. Solutions: renewable energy, plant trees, save electricity.

Stars



The life cycle of stars. What happens after the main-sequence stage depends on the star's mass.

- **Birth:** nebula (gas and dust) → gravity pulls it together → protostar → **nuclear fusion** starts (hydrogen → helium, gives light and heat).
- **Life:** main-sequence star – stable (gravity in = gas pressure out). The Sun is here (4,6 billion years old, about 5 billion to go). Bigger stars burn faster. Blue = hottest, red = coolest.
- **Death – average star (Sun):** red giant → planetary nebula → white dwarf → black dwarf.
- **Death – massive star:** red supergiant → **supernova** → neutron star or **black hole** (light cannot escape).